

Real-Time Collaborative Workspace Agile Management Tool

Project Overview

This project is a full-stack MERN application inspired by Trello/Jira that enables teams to collaborate in real time using workspaces, boards, Kanban task management, chat, invitations, notifications and live collaboration.

How We Started

We began by designing the project architecture, setting up separate client and server applications, connecting MongoDB, implementing JWT authentication, and progressively building CRUD modules before integrating real-time Socket.IO features.

Objectives

- Build a secure collaboration platform.
- Manage workspaces, boards, lists and cards.
- Enable real-time collaboration.
- Provide scalable REST APIs.

Tech Stack

Frontend: React.js, React Router, Zustand, Axios, Socket.IO Client.

Backend: Node.js, Express.js, Socket.IO.

Database: MongoDB with Mongoose.

Authentication: JWT, bcryptjs.

Tools: Git, GitHub, VS Code, Postman.

Modules Implemented

Authentication, Workspace Management, Board Management, Lists, Cards, Drag & Drop, Comments, Notifications, Activity Feed, Search, Typing Indicator, Workspace Invitations, Team Chat, Online Users.

Development Timeline

Week 1: Setup & Authentication.

Week 2: Workspaces & Boards.

Week 3: Lists, Cards & Drag-Drop.

Week 4: Comments, Notifications, Search & Typing Indicator.

Week 5: Workspace Invitations.

Week 6: Team Chat.

Week 7: Online Users.

Challenges Faced

JWT token persistence, Socket.IO synchronization, invitation workflow, duplicate online users, real-time chat synchronization, drag-and-drop consistency, MongoDB schema design, frontend state management.

Results

Successfully developed a real-time collaborative workspace with secure authentication, Kanban workflow, invitations, notifications, chat, online presence and live collaboration.

Discussion

Socket.IO significantly improved collaboration by eliminating page refreshes. Zustand simplified state management while MongoDB provided flexible document storage.

Keywords

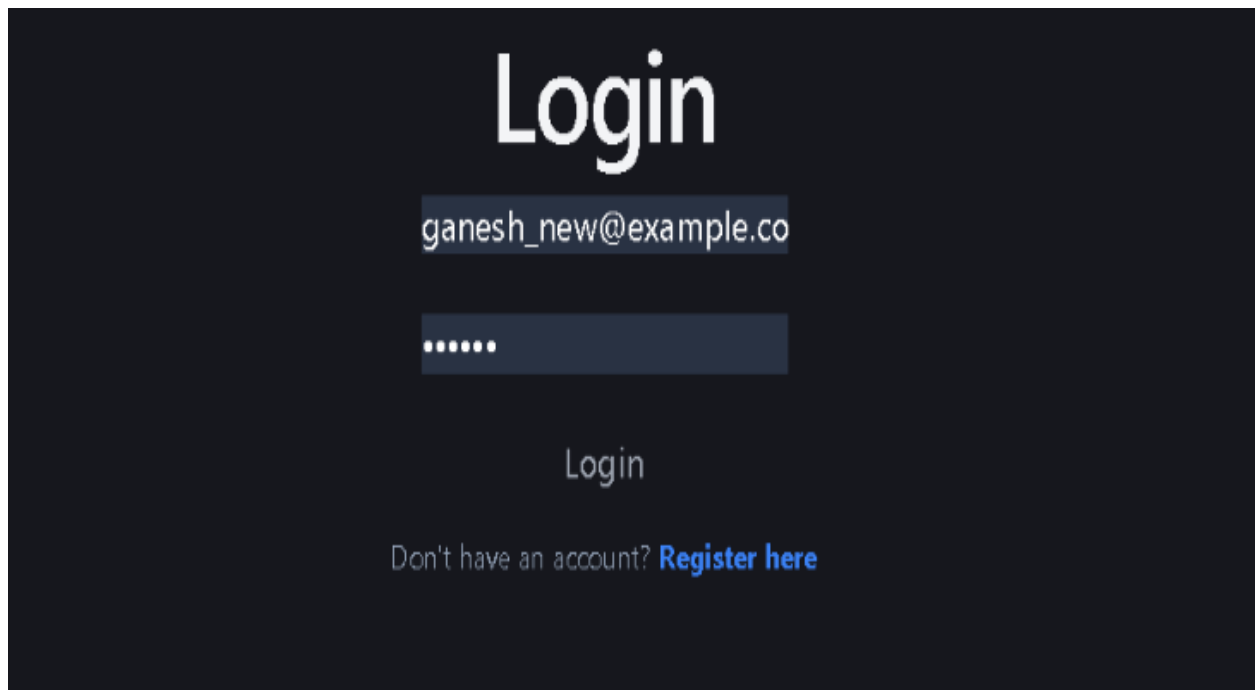
MERN Stack, React, Node.js, Express.js, MongoDB, Socket.IO, JWT, Agile, Kanban, REST API, Real-Time Collaboration, Zustand, Notifications, Chat, Online Users.

Conclusion

The project demonstrates end-to-end full-stack development, secure authentication, REST API development, NoSQL database integration, and real-time collaboration. It is suitable as a portfolio project and showcases software engineering best practices.

Project Outputs / Screenshots

Login Page



Register Page

Register

Name

ganesh_new@example.co

.....

Register

Already have an account? [Login](#)

Dashboard

Workspace Dashboard

[Logout](#)

Total Workspaces

5

Total Members

6

Active Workspaces

5

Search Workspace...

Workspace Dashboard

Workspace Name

Create Workspace

Pending Invitations

My First Workspace

Members: 1

Frontend Test Workspace

Members: 1

Dashboard Workspace

Members: 1

Current User Workspace

Members: 1

Project Alpha

Members: 2

Workspace Details

Workspace Details

Project Alpha

Owner: ganesh_new@example.com

Total Members: 2

Invite Member

Select User

Invite

Boards

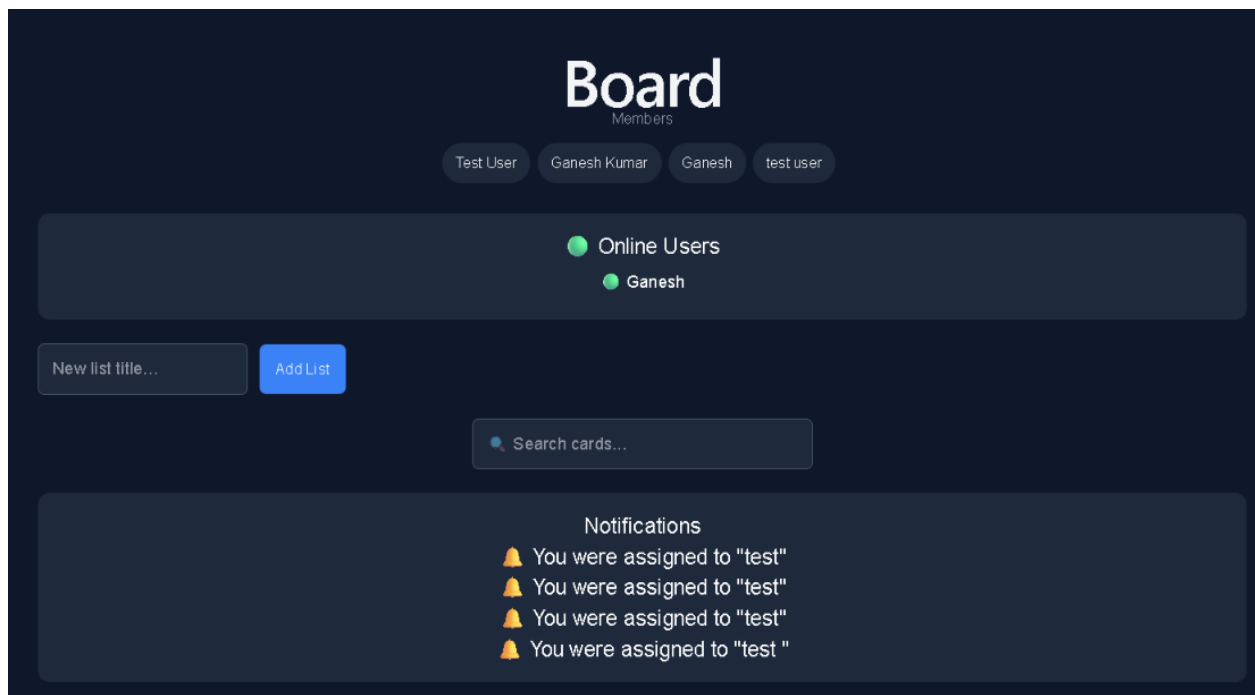
Board Name

Create Board

Development Board

Delete

Board Overview



Team Chat & Kanban Board

